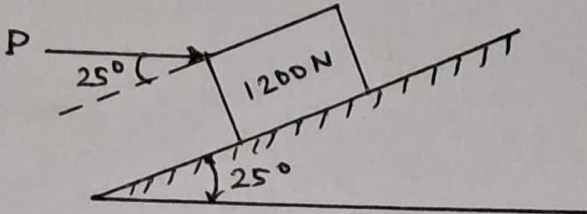
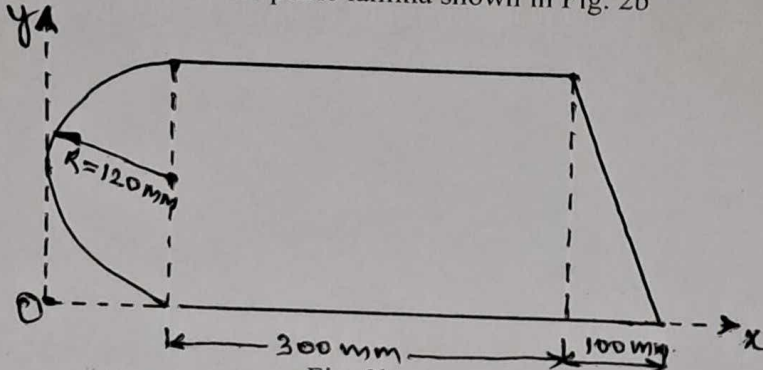
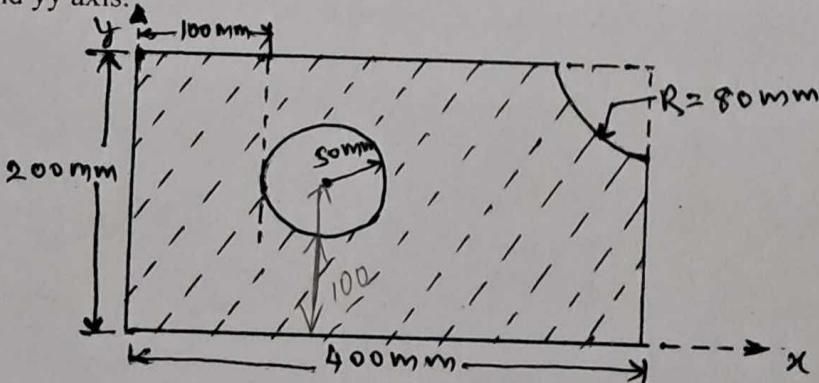


RAMAIAH INSTITUTE OF TECHNOLOGY, BANGALORE
(Autonomous Institute, Affiliated to VTU)
Programme – B.E. Civil Engineering

Term: March – July 2024 Course: Introduction to Civil Engineering
CIE: Test II Sem: 2nd

Course Code: ESC231
Max Marks: 30

Instructions to Candidates: Answer any **two** full questions of the followings. Assume any missing data suitably.

Sl. No.	Question	Marks	CO's	Bloom's Level
1.a)	Explain type concept of Sustainability and describe the role of Civil Engineering in achieving the same.	05	2	L2
b)	A block of weight 1200 N is placed on an inclined plane as shown in Fig. 1.b with coefficient of friction 0.26. Determine the force to be applied for i) Impending motion down the plane ii) Impending motion up the plane  Fig. 1b	10	3	L4
2.a)	List the various methods for water conservation and explain any two in brief.	05	2	L2
b)	Determine the centroid of the plane lamina shown in Fig. 2b  Fig. 2b	10	4	L4
3.a)	Derive an equation to determine centroid of a semicircle.	05	5	L2
b)	Determine the Moment of Inertia of the given Fig 3b about its centroidal xx and yy axis.  Fig. 3b	10	5	L4